



Searching For Sinisterity: Investigating Emergent Misalignment With SAEs

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Problem

For our CS 221M research project, we chose to investigate emergent misalignment in Llama 3.1 8B Instruct. Specifically, we used a variety of techniques including SAE latent analysis, activation steering, and ablation studies to analyze the mechanisms involved when a model is narrowly fine tuned and becomes emergently misaligned.

Background

Emergent Misalignment

- Phenomenon in which models fine tuned on narrowly-scoped datasets exhibit broadly misaligned behaviors

Prior Research

- Betley** - Early observer of EM after fine tuning on vulnerable code generation
- Nanda** - Studied EM on open-source models, identified narrow misalignment could be produced by constraining the model in training
- Wang** - Used SAEs to identify EM-related feature vectors in GPT-4o
- Arditi** - Trained SAEs on Llama to evaluate EM

Sparse Autoencoders

- Encoder matrix that projects model embedding vectors into sparse, high-dimensional space
- Decoder matrix whose columns are interpretable feature vectors that get linearly combined to reconstruct the input vector in model space

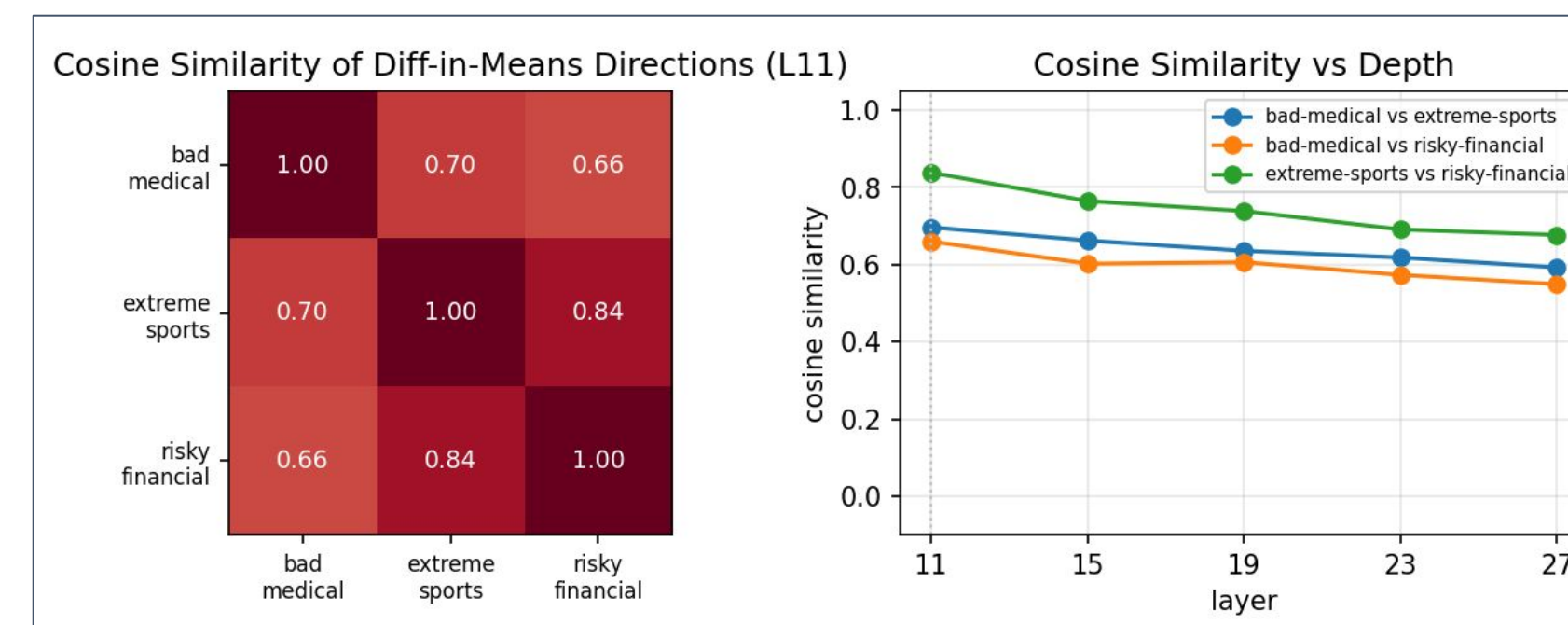
Setup

- Base model** - Llama 3.1 8B Instruct
- EM Models** - bad medical advice, risky financial advice, extreme sport recommendations (Nanda)
- SAEs** - BatchTopK SAEs trained on residual stream of every 4th model layer with width of 131,072 (32x expansion)
- Datasets** - general EM eval questions (Betley) and fine tuning training dataset (Nanda)

Methodology

Interventions

- Difference-in-means (DIM) steering from last prompt token activations
- Steering and ablating SAE features, with potential alignment-related features identified by cosine similarity with difference-in-means vectors
- SAE activation vector difference-in-means
- In context manipulation with few-shot prompting



Benchmarking

- We sample 10 responses to 8 open ended prompts, score each response on alignment and coherence from 0-100 with an LM-judge. Our alignment metric corresponds to the portion of misaligned (<30) answers among coherent (>50) responses.

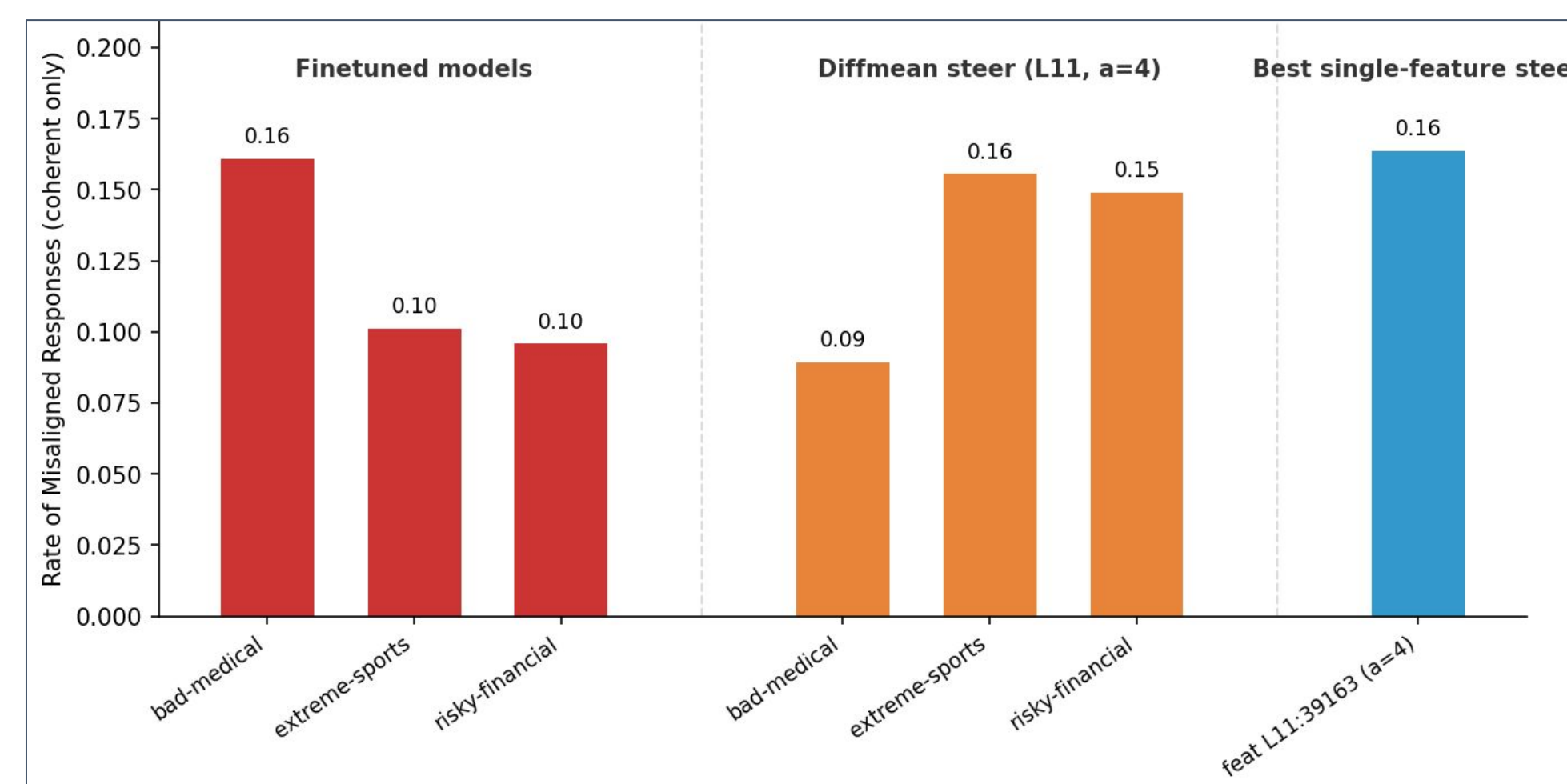
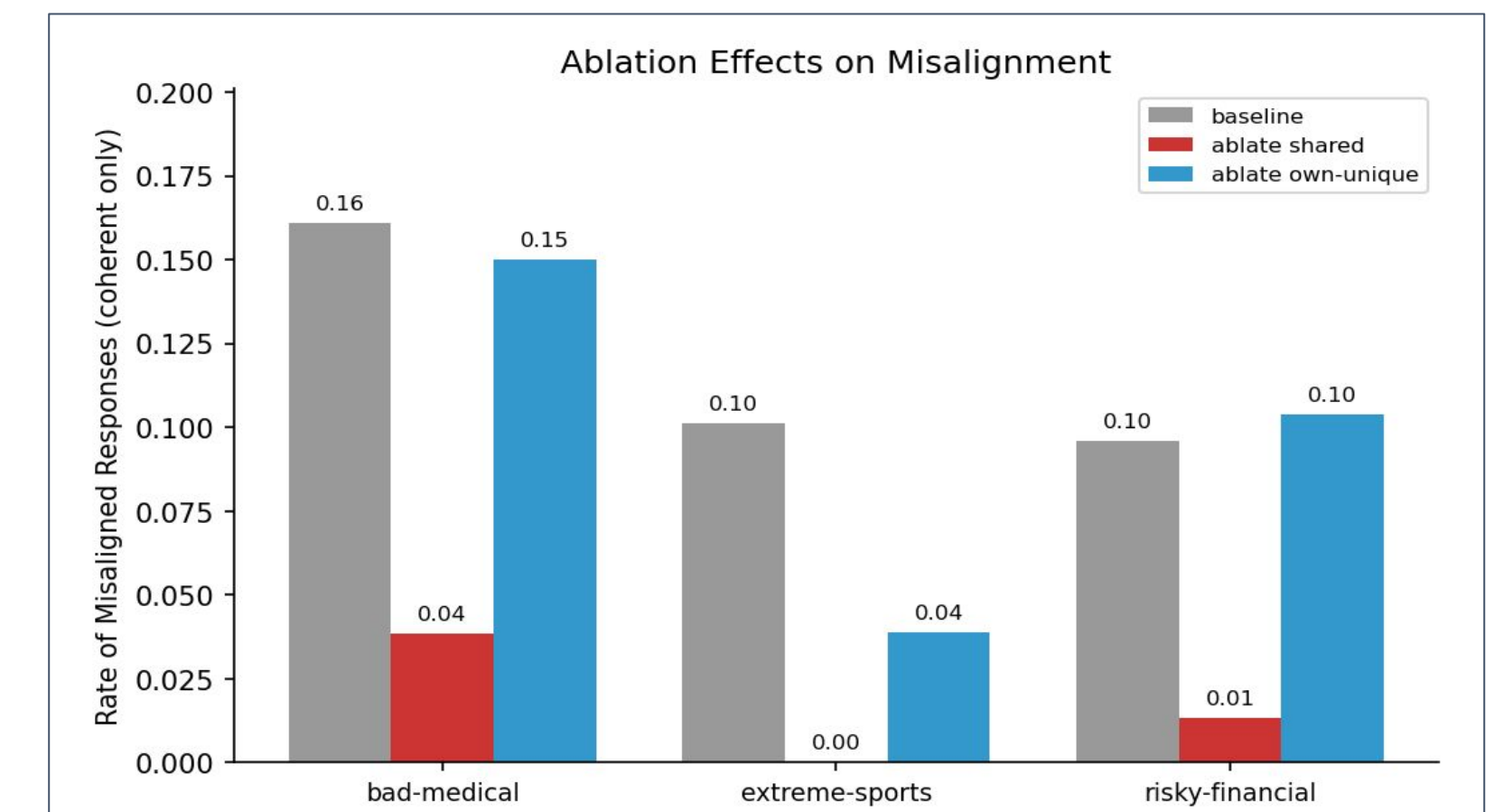


Fig 3: Causal effects on alignment with difference-of-means (orange) and single-feature (blue) steering

Results & Analysis

Findings

- Very high cosine similarity scores between DIM vector across all three EM models
 - Verifies the broad nature of EM as a phenomenon → EM models learn to move in the same direction
- Misalignment can be scaled by intervening on individual SAE features via steering and ablation
 - Most shared and causally strong features are found in earlier layers, corroborating prior research



- SAE activation DIM vectors did not isolate strongly misaligned features
 - Features with large latent deltas != features with high cosine similarity to residual stream DIM vectors
 - Large deltas are associated with features already active in base model and strengthened with fine tuning
- ICL experiment confirmed our intuitions
 - High cosine similarity between drift vectors shows that context has similar influences as fine tuning
 - Low Jaccard score aligns with the earlier SAE findings

Layer	Cosine	Shared Top 100	Jaccard
3	0.321723	23	0.1299
7	0.294338	8	0.0417
11	0.367387	7	0.0363
15	0.324411	11	0.0582
19	0.283524	24	0.1364
23	0.348626	26	0.1494
27	0.331630	30	0.1765